Internship Report

Design and Optimization of a Bidirectional CLLLC Resonant Converter

for a 250 kW *Battery-to-Battery DC–DC Application*

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1 Introduction

The rapid growth of battery energy storage systems and fast-charging infrastructure for electric vehicles creates a strong demand for highly efficient, high-power DC–DC converters capable of bidirectional energy transfer between two DC buses. In many stationary and vehicular applications, two battery banks with nominally equal voltage levels must exchange large amounts of energy, often exceeding 100 kW, without an intermediate AC link.

The **CLLLC resonant converter** has emerged as a preferred topology for such applications. By placing symmetric LC resonant tanks on both sides of a high-frequency transformer, the converter achieves near-sinusoidal current waveforms that naturally support Zero-Voltage Switching (ZVS) of all active devices over a wide operating range. These properties simultaneously reduce switching losses, lower electromagnetic interference (EMI), and allow compact magnetics at frequencies in the tens of kilohertz range. The symmetric structure further ensures identical converter behavior in both power-flow directions.

This report documents the modelling, loss analysis, and numerical parameter optimisation of a 250 kW CLLLC converter intended for a 800 V/800 V battery-to-battery application, carried out during an internship. An exact time-domain state-space model is used for the half-period waveform calculation, avoiding the approximations of fundamental-harmonic analysis (FHA). The two free design parameters — magnetising inductance L_m and resonant inductance L_r (with the resonant capacitance C_r determined by a fixed resonant frequency) — are found by minimising total converter losses subject to ZVS and flux-saturation constraints using gradient-based constrained optimisation (`fmincon`).

Report structure: Section 2 introduces the converter topology and state-space model. Section 3 describes the steady-state solution procedure. Sections 4 and 5 detail current extraction. Section 6 discusses ZVS and residual switching losses. Section 7 presents the system specifications. Section 8 models all loss mechanisms. Section 9 formulates and solves the optimisation problem. Section 10 tabulates all design parameters. Section 11 presents results, and Section 13 outlines future work.

2 Converter Topology and State-Space Model

2.1 Circuit Topology

The CLLLC converter consists of two H-bridge switch networks connected through a symmetric resonant network and a high-frequency transformer. On the primary side, a series resonant inductance L_{r1} and series resonant capacitance C_{r1} form the first LC tank. The transformer magnetising inductance L_m provides energy storage for ZVS. On the secondary side, an identical tank L_{r2}–C_{r2} mirrors the primary structure, making the converter intrinsically bidirectional. For a symmetric design:

$$L_{r1} = L_{r2} = L_r, \qquad C_{r1} = C_{r2} = C_r, \qquad n = 1. \tag{1}$$

The only control variable is the phase shift ϕ between the primary and secondary gating signals. Advancing the secondary gate signals causes power to flow from primary to secondary; reversing the sign of ϕ reverses the power direction.

Figure Placeholder

Insert CLLLC circuit schematic here
(Primary H-bridge → L_{r1}–C_{r1} → Transformer (L_m) → L_{r2}–C_{r2} → Secondary H-bridge)

Figure 1: Bidirectional CLLLC resonant converter with symmetric resonant tanks and high-frequency transformer (turns ratio $n = 1$).

2.2 State Variables

The four independent energy-storage elements give a fourth-order state vector:

$$\boldsymbol{x}(t) = \begin{bmatrix} i_{L_{r1}}(t) \\ i_{L_{r2}}(t) \\ v_{C_{r1}}(t) \\ v_{C_{r2}}(t) \end{bmatrix}. \tag{2}$$

The magnetising current is recovered algebraically as

$$i_{L_m}(t) = i_{L_{r1}}(t) - i_{L_{r2}}(t). \tag{3}$$

2.3 State-Space Equations

Within each piecewise-constant interval the full-bridge voltages applied to the resonant tanks take the values $u_1 = \pm U_1$ and $u_2 = \pm U_2$. Applying KVL and KCL to the CLLLC network yields the linear time-invariant system:

$$\dot{\boldsymbol{x}}(t) = \boldsymbol{A}\,\boldsymbol{x}(t) + \boldsymbol{B}\,\boldsymbol{u}(t), \tag{4}$$

where $\boldsymbol{u} = [u_1,\ u_2]^{\mathsf{T}}$ is the input vector. Defining $\Delta = L_r L_r + L_r L_m + L_r L_m = L_r^2 + 2L_r L_m$ (the determinant of the coupled inductance matrix), the system matrix is:

$$\boldsymbol{A} = \frac{1}{\Delta}\begin{bmatrix} -R_1(L_r + L_m) & -R_2 L_m & -(L_r + L_m) & -L_m \\ -R_1 L_m & -R_2(L_r + L_m) & -L_m & -(L_r + L_m) \\ \Delta/C_r & 0 & 0 & 0 \\ 0 & \Delta/C_r & 0 & 0 \end{bmatrix}, \tag{5}$$

and the input matrix is:

$$\boldsymbol{B} = \frac{1}{\Delta}\begin{bmatrix} (L_r + L_m) & -L_m \\ L_m & -(L_r + L_m) \\ 0 & 0 \\ 0 & 0 \end{bmatrix}. \tag{6}$$

3 Steady-State Solution

3.1 Half-Period Structure

Exploiting half-wave symmetry, the full period $T_s = 1/f_s$ is split into two intervals. Let ϕ be the phase shift in radians ($0 \le \phi \le \pi/2$):

$$T_1 = \frac{T_s}{2} - \frac{\phi}{\omega_s}, \qquad T_2 = \frac{\phi}{\omega_s}, \qquad \omega_s = 2\pi f_s. \tag{7}$$

During interval 1 ($0 \le t \le T_1$) the input vector is $\boldsymbol{u}_1 = [U_1,\ U_2]^{\mathsf{T}}$, and during interval 2 ($T_1 \le t \le T_1 + T_2$) it is $\boldsymbol{u}_2 = [U_1,\ -U_2]^{\mathsf{T}}$.

3.2 Matrix Exponential Propagation

The solution to (4) over a constant-input interval of duration T is:

$$\boldsymbol{x}(T) = e^{\boldsymbol{A}T}\,\boldsymbol{x}(0) + \boldsymbol{A}^{-1}\left(e^{\boldsymbol{A}T} - \boldsymbol{I}\right)\boldsymbol{B}\,\boldsymbol{u}. \tag{8}$$

Define the state transition matrices and forced-response vectors for each interval:

$$\boldsymbol{\Phi}_i = e^{\boldsymbol{A}T_i}, \qquad \boldsymbol{\Gamma}_i = \boldsymbol{A}^{-1}(\boldsymbol{\Phi}_i - \boldsymbol{I})\,\boldsymbol{B}, \quad i = 1, 2. \tag{9}$$

3.3 Periodic Boundary Condition

Half-wave symmetry imposes that the state at $t = T_1 + T_2$ equals the negative of the state at $t = 0$:

$$\boldsymbol{\Phi}_2(\boldsymbol{\Phi}_1\,\boldsymbol{x}_0 + \boldsymbol{\Gamma}_1\,\boldsymbol{u}_1) + \boldsymbol{\Gamma}_2\,\boldsymbol{u}_2 = -\boldsymbol{x}_0. \tag{10}$$

Collecting terms yields a linear system in the unknown initial state $\boldsymbol{x}_0$:

$$\underbrace{(\boldsymbol{\Phi}_2\boldsymbol{\Phi}_1 + \boldsymbol{I})}_{\boldsymbol{M}}\,\boldsymbol{x}_0 = -\boldsymbol{\Phi}_2\,\boldsymbol{\Gamma}_1\,\boldsymbol{u}_1 - \boldsymbol{\Gamma}_2\,\boldsymbol{u}_2 =: \boldsymbol{b}. \tag{11}$$

This 4×4 system is solved directly:

$$\boldsymbol{x}_0 = \boldsymbol{M}^{-1}\boldsymbol{b}. \tag{12}$$

The state at the interval boundary then follows as:

$$\boldsymbol{x}_1 = \boldsymbol{\Phi}_1\,\boldsymbol{x}_0 + \boldsymbol{\Gamma}_1\,\boldsymbol{u}_1. \tag{13}$$

All waveforms over the two sub-intervals are subsequently reconstructed from (8) using $\boldsymbol{x}_0$ and $\boldsymbol{x}_1$ as initial conditions, respectively.

4 Output Current Extraction

The instantaneous secondary-side output current is determined by the direction of i_{Lr2}: during interval 1 it contributes positively to the load, and during interval 2 negatively (the secondary bridge is phase-shifted). The average output current over a full period is:

$$I_{\text{out}} = \frac{2}{T_s}\left[\int_0^{T_1} i_{Lr2}^{(1)}(t)\,\mathrm{d}t - \int_0^{T_2} i_{Lr2}^{(2)}(t)\,\mathrm{d}t\right], \tag{14}$$

where the superscript indicates the interval. In the discretised implementation the integrals are approximated by a Riemann sum over $N = 300$ equally spaced time samples per interval, which is accurate to well within 0.1 % for the smooth waveforms considered here.

For a given set of resonant parameters, the operating phase shift $\phi^\star$ that delivers the nominal reference current

$$I_{\text{ref}} = \frac{P_{\text{nom}}}{U_2} \tag{15}$$

is found by solving $I_{\text{out}}(\phi) = I_{\text{ref}}$ with a one-dimensional root-finding algorithm (`fzero` in MATLAB). A preliminary check verifies that I_{ref} is bracketed within $\phi \in (0, \pi/2)$ before the root-finder is invoked.

5 RMS Current Extraction

The RMS values of the primary and secondary resonant inductor currents are required for conduction-loss computation. Using the half-wave symmetry, the full-period RMS values are:

$$I_{Lr1,\text{rms}} = \sqrt{\frac{2}{T_s}\left[\int_0^{T_1}\left(i_{Lr1}^{(1)}\right)^2\mathrm{d}t + \int_0^{T_2}\left(i_{Lr1}^{(2)}\right)^2\mathrm{d}t\right]}, \tag{16}$$

$$I_{Lr2,\text{rms}} = \sqrt{\frac{2}{T_s}\left[\int_0^{T_1}\left(i_{Lr2}^{(1)}\right)^2\mathrm{d}t + \int_0^{T_2}\left(i_{Lr2}^{(2)}\right)^2\mathrm{d}t\right]}. \tag{17}$$

As with the average current, the integrals are evaluated numerically from the pre-computed waveform samples.

6 Zero-Voltage Switching Analysis

6.1 ZVS Principle

Zero-Voltage Switching is achieved when a switch turns on after its body diode (or parallel capacitance) has already clamped the device voltage to zero. This requires that the resonant inductor current flowing through the switch node has the correct polarity at the switching instant to discharge the node capacitance before the gate signal arrives. ZVS eliminates the capacitive turn-on loss $\frac{1}{2}C_{\text{oss}}V_{\text{ds}}^2 f_s$ and dramatically reduces $\mathrm{d}v/\mathrm{d}t$-induced EMI.

6.2 ZVS at $t = t_0$ — Primary Switches

At $t = t_0$ (the beginning of interval 1) the primary H-bridge commutates. For the turning-on primary switches to achieve ZVS, the primary resonant current must be negative at this instant, i.e. it must flow through the anti-parallel body diodes before the gate pulse arrives:

$$i_{L_{r1}}(t_0) = x_0^{(1)} < 0. \tag{18}$$

This condition is enforced as an inequality constraint in the optimisation.

6.3 ZVS at $t = t_1$ — Secondary Switches

At $t = t_1 = T_1$ the secondary H-bridge commutates. For the turning-on secondary switches to achieve ZVS, the secondary resonant current must be positive:

$$i_{L_{r2}}(t_1) = x_1^{(2)} > 0. \tag{19}$$

6.4 Residual Switching Losses of the Turning-Off Switches

Even with ZVS, the switches that *turn off* still incur switching losses. During turn-off, the device current decreases from its on-state value to zero over a finite fall time. A simplified linear current ramp model gives a turn-off energy proportional to the commutated current. The rise and fall times of the SiC MOSFET are estimated from the device capacitances and gate-drive circuit parameters:

$$t_r = \frac{V_{\text{ds}}\, C_{\text{rss}}}{V_p - V_{\text{off}}}\, R_g, \tag{20}$$

$$t_f = R_g\, C_{\text{iss}} \ln\!\left(\frac{V_p - V_{\text{off}}}{V_{\text{th}} - V_{\text{off}}}\right). \tag{21}$$

The total switching loss per bridge (one commutation event per half-period) is then modelled as:

$$P_{\text{sw}} = \tfrac{1}{2}\, U_1\, i_{\text{sw}}\, (t_r + t_f)\, f_s, \tag{22}$$

where i_{sw} is the magnitude of the inductor current at the relevant switching instant: $|x_0^{(1)}|$ for the primary bridge and $|x_1^{(2)}|$ for the secondary bridge.

7 System Specifications

The converter interfaces two lithium-ion battery banks within a stationary energy-storage system. Both banks have a nominal voltage of $U_1 = U_2 = 800\,\text{V}$, and the converter is rated for a nominal bidirectional power transfer of $P_{\text{nom}} = 250\,\text{kW}$. This yields a nominal output current of

$$I_{\text{ref}} = \frac{P_{\text{nom}}}{U_2} = \frac{250\,000}{800} = 312.5\,\text{A}. \tag{23}$$

The switching frequency is chosen as $f_s = 12\,\text{kHz}$, balancing magnetics volume against switching losses for SiC MOSFET technology. The resonant frequency is fixed above the switching frequency to ensure that the converter operates in the inductive region, which is a prerequisite for ZVS:

$$f_r = 1.2\, f_s = 14.4\,\text{kHz}, \qquad \omega_r = 2\pi f_r. \tag{24}$$

For a symmetric design with equal primary and secondary resonant elements, the resonant frequency sets the relationship between L_r and C_r:

$$C_r = \frac{1}{\omega_r^2 L_r}. \tag{25}$$

This constraint reduces the resonant tank design to a single free parameter, L_r, discussed further in Section 9.

The SiC half-bridge modules selected are the Wolfspeed C3M0016120U2, a 1200 V, 16 mΩ MOSFET optimised for hard- and soft-switched applications. Key device parameters used in the switching loss model are listed in Section 10.

8 Loss Modelling

The total converter loss function minimised during optimisation comprises three contributions:

$$P_{\text{total}}(L_m, L_r) = P_{\text{cond}} + P_{\text{sw}} + P_{\text{core}}. \tag{26}$$

8.1 Conduction Losses

Winding resistances R_1 (primary) and R_2 (secondary) are assumed constant. The conduction losses are:

$$P_{\text{cond}} = I_{Lr1,\text{rms}}^2 R_1 + I_{Lr2,\text{rms}}^2 R_2. \tag{27}$$

8.2 Switching Losses

Using the model from Section 6, the total switching loss from both bridges is:

$$P_{\text{sw}} = \underbrace{\tfrac{1}{2} U_1 \left|x_0^{(1)}\right| (t_r + t_f) f_s}_{\text{primary bridge}} + \underbrace{\tfrac{1}{2} U_1 \left|x_1^{(2)}\right| (t_r + t_f) f_s}_{\text{secondary bridge}}. \tag{28}$$

All four switches in each bridge turn off in pairs; the model accounts for one effective commutation event per bridge per half-period.

8.3 Core (Transformer) Losses

Core losses are modelled with the Steinmetz equation. The peak magnetising current in the transformer core is:

$$\hat{i}_{L_m} = \max_t \left| i_{L_m}(t) \right|, \tag{29}$$

from which the peak flux density follows:

$$\hat{B} = \frac{L_m \, \hat{i}_{L_m}}{N_1 \, A_c}, \tag{30}$$

where N_1 is the number of primary turns and A_c the core cross-sectional area. The Steinmetz core loss is:

$$P_{\text{core}} = k_s \, f_s^{\,\alpha} \, \hat{B}^{\,\beta} \, V_c, \tag{31}$$

where k_s, α, and β are material-dependent Steinmetz coefficients and V_c is the total core volume.

Note on the winding resistance model: In the current implementation, R_1 and R_2 are treated as frequency-independent DC resistances. This is a simplification; in reality, skin and proximity effects increase the effective AC resistance at $f_s = 12\,\text{kHz}$ and must be accounted for in a refined design (see Section 13).

Note on the core loss model: The Steinmetz model assumes sinusoidal flux excitation and does not capture the non-sinusoidal waveforms present at the converter operating point. A more accurate core loss prediction could be obtained with the improved Generalized Steinmetz Equation (iGSE), as discussed in Section 13. Furthermore, the waveform simulation itself assumes linear magnetics and does not feed back the core loss into the electrical waveforms.

9 Optimisation Problem

9.1 Design Variables

As established in Section 7, the resonant frequency constraint (25) ties C_r to L_r, leaving two independent design parameters:

$$\boldsymbol{p} = \begin{bmatrix} L_m \\ L_r \end{bmatrix}. \tag{32}$$

The resonant impedance $Z_r = \sqrt{L_r/C_r} = \omega_r L_r$ scales together with L_r; varying L_r is therefore equivalent to varying Z_r.

9.2 Objective Function

The objective is to minimise total converter losses at nominal power:

$$\min_{\boldsymbol{p}} \; J(\boldsymbol{p}) = P_{\text{cond}}(\boldsymbol{p}) + P_{\text{sw}}(\boldsymbol{p}) + P_{\text{core}}(\boldsymbol{p}), \tag{33}$$

where for each candidate $\boldsymbol{p}$ the operating phase shift $\phi^\star(\boldsymbol{p})$ is first found by solving $I_{\text{out}}(\phi;\, \boldsymbol{p}) = I_{\text{ref}}$, and all loss terms are then evaluated at $\phi^\star$.

9.3 Constraints

The optimisation is subject to the following inequality constraints:

$$g_1: \quad x_0^{(1)} \le 0 \qquad \text{(ZVS, primary switches at } t_0\text{)} \tag{34}$$

$$g_2: \quad x_1^{(2)} \ge 0 \qquad \text{(ZVS, secondary switches at } t_1\text{)} \tag{35}$$

$$g_3: \quad \phi^\star \ge 0 \qquad \text{(positive phase shift)} \tag{36}$$

$$g_4: \quad \phi^\star \le \tfrac{\pi}{2} \qquad \text{(phase shift upper bound)} \tag{37}$$

$$g_5: \quad \hat{B} \le B_{\text{sat}} = 0.30\,\text{T} \qquad \text{(no core saturation)} \tag{38}$$

$$g_6: \quad L_m \ge 2\,L_r \qquad \text{(physical inductance ratio)} \tag{39}$$

The inductance ratio constraint (39) ensures that the magnetising inductance is large compared to the resonant inductance ($k = L_m/L_r \ge 2$), which is a practical requirement for a well-coupled transformer design. Box constraints on the search space are:

$$L_m \in [50\,\mu\text{H},\; 300\,\mu\text{H}], \qquad L_r \in [1\,\mu\text{H},\; 30\,\mu\text{H}]. \tag{40}$$

9.4 Algorithm

The optimisation uses MATLAB's `fmincon` with the interior-point algorithm, which handles nonlinear inequality constraints natively. At each iteration:

1. The state matrices $\boldsymbol{A}(\boldsymbol{p})$ and $\boldsymbol{B}(\boldsymbol{p})$ are assembled.
2. The periodic initial state $\boldsymbol{x}_0$ is solved via (12).
3. The root $\phi^\star$ is found with `fzero`.
4. All waveforms are sampled, and losses (26) are computed.
5. Constraint values g_1–g_6 are returned to the solver.

Optimality and constraint tolerances are both set to 10^{-6}.

10 Design Parameters

Table 1 summarises all fixed system, device, core, and constraint parameters used in the analysis.

Table 1: Summary of converter design and optimisation parameters.

Parameter	Symbol	Value	Unit
System			
DC bus voltage (both sides)	U_1, U_2	800	V
Nominal output power	P_{nom}	250 000	W
Switching frequency	f_s	12 000	Hz
Resonant frequency	f_r	14 400	Hz
Winding resistance (each side)	R_1, R_2	0.026	Ω
Switch — Wolfspeed C3M0016120U2			
Reverse transfer capacitance	C_{rss}	11	pF
Input capacitance	C_{iss}	6 000	pF
Drain–source voltage	V_{ds}	800	V
Gate drive voltage	V_p	7	V
Gate off voltage	V_{off}	0	V
Gate drive resistance	R_g	10	Ω
Threshold voltage	V_{th}	2.1	V
Transformer core (Steinmetz model)			
Core volume (total)	V_c	200	cm^3
Core cross-section	A_c	3.68×10^{-4}	m^2
Primary turns	N_1	150	—
Steinmetz coefficient	k_s	42 366	—
Frequency exponent	α	1.160	—
Flux density exponent	β	2.800	—
Optimisation bounds and constraints			
Saturation flux density	B_{sat}	0.30	T
Inductance ratio minimum	L_m/L_r	≥ 2	—
L_m search range	L_m	50 – 300	μH
L_r search range	L_r	1 – 30	μH

11 Results

11.1 Optimised Parameters

Table 2 lists the optimal design parameters found by `fmincon` together with the resulting loss breakdown and converter efficiency at nominal power.

Table 2: Optimisation results at nominal operating point ($P_{\text{nom}} = 250\,\text{kW}$).

Quantity	**Symbol**	**Value**	**Unit**
Optimal magnetising inductance	$L_m^\star$	[insert]	μH
Optimal resonant inductance	$L_r^\star$	[insert]	μH
Optimal resonant capacitance	$C_r^\star$	[insert]	nF
Inductance ratio	$L_m^\star / L_r^\star$	[insert]	—
Operating phase shift	$\phi^\star$	[insert]	deg
Conduction losses	P_{cond}	[insert]	W
Switching losses	P_{sw}	[insert]	W
Core losses	P_{core}	[insert]	W
Total losses	$J^\star$	[insert]	W
Efficiency	η	[insert]	%
ZVS check: primary ($x_0^{(1)} < 0$)	—	[insert]	A
ZVS check: secondary ($x_1^{(2)} > 0$)	—	[insert]	A
Peak flux density	$\hat{B}$	[insert]	T

11.2 Waveforms at Nominal Operating Point

Figure 2 shows the output current as a function of phase shift for the optimised converter, with the operating point marked. Figures 3 and 4 show the inductor current and capacitor voltage waveforms over one half-period.

Figure Placeholder

I_{out} vs. phase shift ϕ — insert plot of output current characteristic with I_{ref} and $\phi^\star$ marked

Figure 2: Output current I_{out} as a function of phase shift ϕ. The dashed red line indicates I_{ref} and the green line marks the optimal operating point $\phi^\star$.

Figure Placeholder

Inductor current waveforms ($i_{L_{r1}}$, $i_{L_{r2}}$, i_{L_m}) over one half-period — insert MATLAB plot

Figure 3: Resonant inductor currents $i_{L_{r1}}$ (blue), $i_{L_{r2}}$ (red), and magnetising current i_{L_m} (black) over one half-period at $\phi = \phi^\star$. Solid lines: interval 1; dashed lines: interval 2.

Figure Placeholder

Capacitor voltage waveforms ($v_{C_{r1}}$, $v_{C_{r2}}$) over one half-period — insert MATLAB plot

Figure 4: Resonant capacitor voltages $v_{C_{r1}}$ (blue) and $v_{C_{r2}}$ (red) over one half-period at $\phi = \phi^\star$. Solid lines: interval 1; dashed lines: interval 2.

12 Conclusions

A complete analytical framework for the steady-state analysis and loss-optimal design of a 250 kW CLLLC resonant converter has been presented. The exact time-domain

state-space model correctly resolves the periodic waveforms without FHA approximations, enabling accurate computation of RMS currents, switching-instant conditions, and peak flux density. The `fmincon`-based optimisation successfully minimises the combined conduction, switching, and core losses while enforcing ZVS on all active bridges and preventing core saturation.

The results demonstrate that the CLLLC topology is well-suited to the target application, achieving efficient bidirectional power transfer between equal 800 V bus voltages with a compact resonant tank. The methodology developed here provides a solid and extensible foundation for further refinement of the converter design.

13 Future Work

The current design framework provides a good starting point, but several important refinements and extensions should be addressed before a hardware prototype is built.

1 — Simulation Validation

The analytical model has not yet been verified against a circuit simulation. A time-domain simulation in PLECS, MATLAB/Simulink, or LTspice should be used to validate the steady-state waveforms, the ZVS behaviour at the switching instants, and the accuracy of the loss model before hardware construction.

2 — Improved Core Design

The transformer core is currently characterised by placeholder geometry parameters. A dedicated core design study should select a specific core geometry (e.g. a medium-frequency E-core or toroid), determine the number of turns from flux density and window utilisation constraints, and confirm that the required inductances can be realised within the chosen geometry.

3 — Frequency-Dependent Winding Resistance

The winding resistances R_1 and R_2 are presently modelled as DC values. At $f_s = 12\,\text{kHz}$, skin and proximity effects significantly increase the effective AC resistance, especially for the large-cross-section conductors required at 250 kW. Litz wire winding analysis or FEM simulations should replace the constant resistance assumption.

4 — Core Loss Feedback into Waveform Model

The iGSE (improved Generalized Steinmetz Equation) should replace the basic Steinmetz model to correctly capture non-sinusoidal flux excitation. More fundamentally, the waveform simulation currently assumes ideal linear magnetics; a magnetics model that reflects nonlinear saturation behaviour and feeds core losses back into the electrical circuit would improve accuracy near the saturation boundary.

5 — Joint Optimisation of f_r and f_s

In the current study, the resonant and switching frequencies are fixed. A broader optimisation that simultaneously varies f_r, f_s, and the ratio f_r/f_s would likely reveal a better global operating point. Lower f_s reduces switching losses but increases magnetics volume; the trade-off is strongly loss-model-dependent.

6 — Three-Level Modulation for RMS Reduction

The present single-phase-shift (SPS) modulation scheme has only one degree of freedom. Extended phase-shift (EPS) or dual-phase-shift (DPS) modulation introduces an additional duty-cycle variable per bridge, and three-level modulation of each bridge further expands the control space. These advanced modulation strategies can significantly reduce the RMS current and thus conduction losses at partial load, at the cost of increased control complexity.

7 — Partial-Load Efficiency Optimisation

The current optimisation targets a single nominal operating point. A weighted loss objective over a representative drive-cycle load profile would produce a design better suited to real-world deployment, where the converter may spend most operating time at 20–60 % of rated power.

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